Weekly Analysis Report:

Log Analysis using Splunk (Day 1–4):

Objective:

The objective of the project is to understand the **log analysis** using **SIEM tool Splunk** and to identify the **suspicious login activities** such as **failed logins** and **brute-force attacks**.

Tools and Data used:

Tools: Splunk

Datasets: log_sample_1.csv Day 1 2

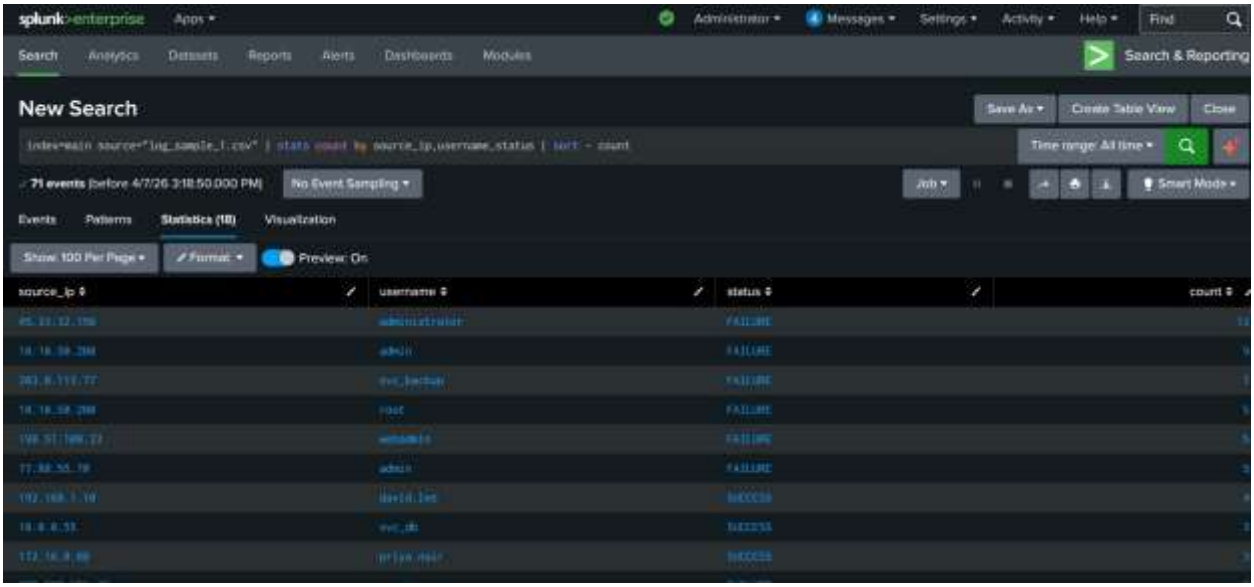
windows_events.csv Day 3 4

Methodology :

day 1-2: Log Analysis

- Imported and analyzed log_sample_1.csv.
- Identified key fields: timestamp ,source Ip, status, username.
- Sorted logs manually.
- Performed manual filtering for specific Ip's.

Log data visualization in Splunk showing login events

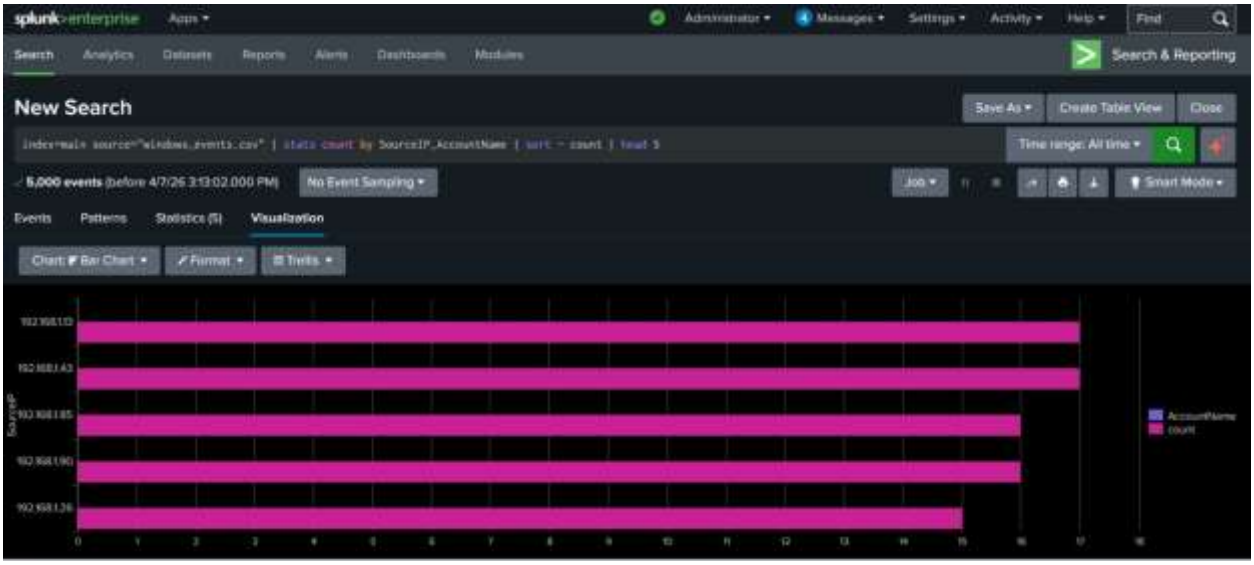


Finally in Day 1-2 analysis we found that some Ip's are effected by brute-force but those are failed logins.

day 3-4: Windows Event Log Analysis

- Imported windows_events.csv to Splunk.
- Focused on Event ID 4625 Failed Login).
- Identified Top 5 source Ip's and targeted usernames.

Most Targeted Ip’s and Usernames



Source IP	Username	Failed Attempts
192.168.1.26	john	13
192.168.1.85	administrator	12
192.168.1.16	svc_app	11
192.168.1.21	svc_app	11
192.168.1.62	guest	11

From day 3 4 analysis we found that the top 5 Ip’s and usernames have been affected.

Summary of findings:

In day 1-2 Logs contains both successful logins and failure logins in the data.

In day 3 4 windows event log analysis contains the Event ID of windows By sorting the Ip’s we can see the which Ip’s show high failed attempts.

We can see some usernames are repeatedly targeted in log analysis dataset and windows event log analysis dataset.

Observation:

I have observed multiple failed attempts from specific Ip's(possible brute-force attempts).

Some Ip's only showing Failure status and i doesn't have success status(unsuccessful attack attempts).

Many Ip's have affected to Brute-Force attack but its not succussed so Password hasn't Found.

Also we can see the Normal Behavior which means failure followed by success with the count limit ≤ 2.

Timestamp parsing issue initially caused logs to be invisible until corrected.

Challenges Faced:

- Incorrect timestamp parsing in csv logs.
- Event doesn't visible due to time range issue.
- Required manual correction at some point of searching and reporting.
- Event not visible initially.

Conclusion:

This analysis helped me to identify the Suspicious login pattern and to understand the log analysis and windows event using Splunk and I have learnt the difference between normal and suspicious behavior and analyzed failed login trends effectively.

Recommendations:

Need to enable Multi-factor Authentication MFA.

Can implement account lock-out policy after many failed attempts.

Ensure proper log formatting and time parsing.

Need to monitor the Ip's after the Brute-force attack which has been effected.

Key Points Learned:

- Understanding of Log Structure and field extraction.
- Learnt to write the SPL query and to optimize it.
- Difference between normal login behavior and attack pattern.
- Hands-on Experience in detecting brute-force attempts.